blacklistManagement

Service Description

Service ID: *"blacklistManagement"*

Abstract
This document describes the `blacklistManagement` service, version 1.0.0, produced by the Blacklist support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `blacklistManagement` service, which provides operator management of the blacklist: adding systems with a reason and expiry, querying entries and inactivating them. It is produced by the Blacklist system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

New in Arrowhead 5.

1.2 How This Service Is Meant to Be Used

Operators, the onboarding procedure (off-boarded systems are blacklisted) and security tooling add entries with `create`, inspect them with `query` and lift them with `remove`.

1.3 Important Delimitations and Dependencies

Entries are inactivated, not deleted, so the history is kept. Access follows the management policy of the Blacklist.

2 Service Interface

This section describes the interfaces to the `blacklistManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

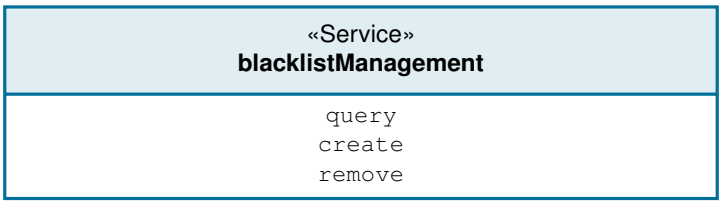


Figure 1: SysML-style block of the `blacklistManagement` service and its operations.

The following interface operations are available.

2.1 operation query (BlacklistQueryRequest) : BlacklistEntryListResponse

Returns a page of entries matching the filter.

2.2 operation create (BlacklistCreateListRequest) : BlacklistEntryListResponse

Adds blacklist entries with reason and optional expiry.

2.3 operation remove (List<Name>)

Inactivates the entries of the named systems.

3 Information Model

Here, all data objects that can be part of the `blacklistManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.7, which are used to represent things like hashes and identifiers.

3.1 struct BlacklistCreateListRequest

Entries to add to the blacklist.

Field	Type	Description
entities	List<BlacklistCreateRequest>	Entries to create.

3.2 struct BlacklistCreateRequest

One system to blacklist.

Field	Type	Description
systemName	Name	System to blacklist.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
reason	String	Reason of the blacklisting.

3.3 struct BlacklistEntry

A blacklist entry.

Field	Type	Description
systemName	Name	The blacklisted system.
createdBy	Name	System that created the entry.
revokedBy	String	System that inactivated the entry, if any.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.
reason	String	Reason of the blacklisting.
expiresAt	DateTime	Expiry of the entry; no value means permanent.
active	Boolean	Whether the entry is in force.

3.4 struct BlacklistEntryListResponse

List of blacklist entries.

Field	Type	Description
entries	List<BlacklistEntry>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.5 struct BlacklistQueryRequest

Paged filter for blacklist entries.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
systemNames	List<Name>	System names to match.
mode	String	Which entries to return: `ALL`, `ACTIVES` or `INACTIVES`.
issuers	List<String>	Systems that created the entries.
revokers	List<String>	Systems that inactivated the entries.
reason	String	Reason of the blacklisting.
alivesAt	DateTime	Only entries in force at this time.

3.6 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.7 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “Blacklist – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/Blacklist/documentation/v5.2.2/Blacklist-v5.2.2-SysD.pdf.

[2] ——, “blacklistManagement HTTP/TLS/JSON – Interface Design Description (Blacklist),” Eclipse Arrowhead, IDD 5.2.2, 2026, blacklist-v5.2.2-blacklistManagement-HTTP-IDD.pdf.

[3] ——, “blacklistManagement MQTT/TLS/JSON – Interface Design Description (Blacklist),” Eclipse Arrowhead, IDD 5.2.2, 2026, blacklist-v5.2.2-blacklistManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	